

Nachiket Rajendra Gadilohar

Software Programmer (Ai Focused)

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Profile summary

AI Engineer with 2+ years of expertise in developing production AI pipelines for engineering automation. Skilled in applied machine learning, computer vision, NLP, OCR, generative AI, and analytical problem-solving. Proven track record in delivering solutions that reduce manual design effort by over 80%, achieving 90%+ model accuracy in industrial environments. Proficient in Retrieval-Augmented Generation (RAG), multi-agent LLM systems, prompt engineering, vector embeddings, and FastAPI-based API development. Experienced in cloud deployment and monitoring on Azure, AWS, and GCP, alongside mentoring junior engineers to enhance team capabilities.

Skills

Python

FastAPI

Machine Learning

Deep Learning

Natural Language Processing

Retrieval-Augmented Generation

Multi-Agent LLM Systems

Vector Embeddings

Knowledge Graphs

Prompt Engineering

OCR

Computer Vision

SQL

REST API Design

Azure

Certifications

- PwC Switzerland — Power BI Job Simulation
- Accenture North America — Data Analytics and Visualization Job Simulation
- Tata Group — Data Visualization: Empowering Business with Effective Insights Job Simulation
- BCG — Data Science Job Simulation
- BCG — Generative AI (GenAI) Job Simulation
- Machine Learning — Fundamentals of Python Machine Learning
- Introduction to Big Data with Spark and Hadoop

Work Experience

Software Programmer (Ai Focused) — Trans Tech Projects Pvt. Ltd.	Apr 2025 - May 2026
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- Engineered an AI-driven AutoCAD automation pipeline leveraging Python and computer vision, reducing design turnaround time by 81% (from ~370 to ~70 hours) while ensuring over 90% accuracy.
- Developed a PDF-to-XML document intelligence system using Python, FastAPI, and OCR technology, achieving 95% data extraction accuracy and eliminating manual data entry in electrical simulation workflows.
- Led requirement analysis, model development, API integration, deployment, and monitoring for multiple engineering automation projects, collaborating closely with engineering stakeholders to deliver effective solutions.
- Mentored two junior team members by creating standardized AI documentation templates and conducting hands-on knowledge transfer sessions to enhance AI project delivery and team proficiency.

ASIC Intern — Personifywy	Aug 2024 - Sep 2024
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- Applied data analysis techniques to real business datasets during a project-based internship while adhering to confidentiality and IP protection standards.

Laboratory Assistant (Summer Internship) — DRDO, Ministry of Defence, Government of India	Mar 2020 - May 2020
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- Assisted in data collection and analysis in a defense research laboratory, operating advanced equipment and collaborating with scientists to ensure accuracy of results under strict safety protocols.

Internships	
— Personify	46 Days

- Applied data analysis techniques to real business datasets during a project-based internship while adhering to confidentiality and IP protection standards.

Projects
paperbrain — AI-Powered PDF Intelligence

- Built a production-grade RAG document intelligence system with a 7-agent processing pipeline (Extract, Analyze, Preprocess, Optimize, Synthesize, Validate, Assemble), FAISS vector search, real-time SSE token streaming, and multi-provider LLM failover.

hateguard-nlp — Production Hate Speech Classifier

- Delivered an end-to-end production NLP classifier via a 6-stage MLOps pipeline (ingestion, validation, transformation, training, evaluation, model push) using LSTM, AWS S3 for artifacts, CircleCI CI/CD, and a Dockerized Flask app deployed on AWS EC2.

tripmind-ai — Multi-Agent Travel Planning System
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- Engineered a multi-agent travel intelligence system orchestrating specialized agents that aggregate real-time data from multiple APIs and evaluated candidate LLMs before deploying via Docker on AWS EC2 with GitHub Actions CI/CD.

Cotton Disease Prediction (Deep Learning)

- Compared transfer-learning architectures (InceptionV3, ResNet50, ResNet152V2) for 4-class cotton disease classification and trained ResNet50 on 1,951 labeled images.

Education

MS/M.Sc(Science) Computers	2025
Symbiosis Institute of Geoinformatics, Pune	
Grade - 6/10	

B.Sc Physics	2022
Fergusson College, Pune	
Marks - 74%	

12th	2019
Maharashtra, English	
Marks - 81%	

10th	2017
Maharashtra, English	
Marks - 84%	

Extra-curricular activities
Mentoring

Mentored two junior team members on AI documentation standards and delivery best practices to improve project handoff quality.

Achievements
Key Achievements

Reduced AutoCAD design turnaround time by 81% and achieved 90%+ model accuracy in industrial deployments;

developed a PDF-to-XML extraction pipeline with **95%** accuracy.

Additional information

Hobbies: Technical Blogging, Open Source Contribution, Competitive Programming

Languages: English